

**In one line:** Since iOS 16 navigation is **data**: a `NavigationStack` renders a **path** (an array of Hashable values) that *you* own; `navigationDestination(for:)` maps each value *type* to a screen. Push = append, pop = remove, deep link = assign a whole path.

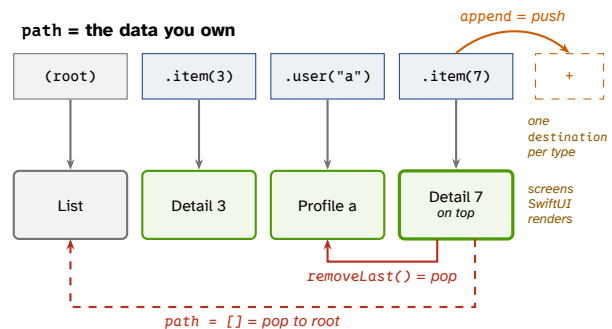
## How it works

- `NavigationStack(path: root:)` (16) — the path is `[Route]` (one type) or `NavigationPath`: a *type-erased* list of Hashable values, so one stack can hold `Item`, `User`, `Int`..., each routed by its own destination.
- `NavigationLink(_:value:)` pushes a *value*, not a view — lazy, decoupled, programmatic. `NavigationLink { Dest() } label: {}` (view-based) still works but cannot be driven from the path.
- `.navigationDestination(for: T.self) { v in ... }` — **one** per type per stack, inside the stack, on a view that is always there (not in a lazy row). iOS 17 adds `navigationDestination(item:)`; `(isPresented:)` is 16.
- **Programmatic:** `push path.append(x)`; `pop path.removeLast()`; `pop-to-root path = []` / `path.removeAll()` for an array, or `path.removeLast(path.count) / path = NavigationPath()` — `NavigationPath` has *no* `removeAll`.
- **Restoration:** if every element is `Codable`, `path.codable` gives a `NavigationPath.CodableRepresentation` (else `nil`); encode to `@SceneStorage` / disk, rebuild with `NavigationPath(rep)`.
- `NavigationSplitView` (16) — 2 columns (sidebar, detail) or 3 (+content); selection drives the next column; `columnVisibility` binding; on compact width it **collapses into a stack** (preferred `CompactColumn`, 17).
- **Modals:** `.sheet` (card, swipe-dismiss, `.presentationDetents([.medium, .large])` 16); `.fullScreenCover` (no swipe-dismiss); `.popover` (bubble on iPad/Mac, a sheet on compact unless `.presentationCompactAdaptation`, 16.4). Each takes `isPresented: (Bool)` or `item: (Identifiable? — preferred when content depends on data)`. Block swipe: `.interactiveDismissDisabled()`.
- `@Environment(\.dismiss)` (15) — pops the pushed view or closes the sheet/cover that presented this view: the *nearest* context.
- **Tabs:** one `NavigationStack` *per tab*, each with its own path.
- **Router:** an `@Observable` object owning path (+ sheet state), put in the environment, so any view calls `router.show(.x)` and `.onOpenURL` maps a URL to routes.

## Example — router + deep link

```
enum Route: Hashable, Codable { case item(Int), user(String) }
@Observable final class Router {
  var path: [Route] = []
  func open(_ url: URL) { path = Route.parse(url) } // all at once
}
struct Root: View {
  @State private var router = Router()
  var body: some View {
    NavigationStack(path: $router.path) {
      List(1..<50, id: \.self) { n in
        NavigationLink("Item \(n)", value: Route.item(n))
        .navigationDestination(for: Route.self) { Screen(route: $0) }
      }
      .environment(router).onOpenURL { router.open($0) }
    }
  }
}
```

## The stack is the path



URL app://item/3/user/a → parse → path = [.item(3), .user("a")]

Deep link = assign the whole array; SwiftUI pushes every screen, back button included. Codable routes ⇒ the same array restores state on relaunch.

## Traps

- ! `NavigationLink(isActive:)` / `(tag:selection:)` are **deprecated** (iOS 16), and so is `NavigationView`: one `Bool` per link, buggy pops, no restoration.
- ! `navigationDestination` inside a `List/LazyVStack` row, or two for one type — “no matching destination” or the wrong screen. Declare once, on a stable ancestor.
- ! `.sheet(isPresented:)` + separate `@State` var selected — can present stale/`nil` data. Use `.sheet(item: $selected)`.
- ! `dismiss()` inside a `NavigationStack` *within* a sheet pops the inner stack, it does not close the sheet — pass a closure/binding or read `dismiss` at sheet level.
- ! `.navigationTitle` goes on the *content*, not on `NavigationStack`; a split view tested only on iPad shows a blank detail when collapsed on iPhone.
- ! Nesting a `NavigationStack` in a pushed screen = two bars / broken back stack.

## Remember

**“Push values, not views.”** Path = `[Hashable]`; destination = *per type*; item: beats `isPresented`; dismiss = nearest context.